

Primal Dual Algorithm — The Framework

Main idea: If (x, p) PD-feasible
and CS holds $\Rightarrow$ optimal.

P-D idea: start with dual feasible p

try to find x to satisfy CS

If we cannot, use this "failure"

to update p to a better

(and always feasible) dual solⁿ p .

→ Update is often an easy subproblem to solve.

Restricted Primal (RP)

start w/ dual f. $p: p^T A \leq c^T$

$J =$ "admissible" columns
 $= \{j: p^T A_j = c_j\}$. $// x_j = 0 \ \forall j \notin J$
for CS to hold.

Primal:

min: $c^T x$

st: $Ax = b$

$x \geq 0$

wlog, assume $b \geq 0$.

(RP) min: $\sum x_i^a$

st: $\sum_{j \in J} a_{ij} x_j + x_i^a = b_i \quad // \quad A_J x_J + x^a = b$

$x_J \geq 0$

$x^a \geq 0$

$x_{J^c} = 0 /$

If obj. = 0 then stop -

we have found x that satisfies

CS with p .

Dual Restricted Primal (DRP)

$$\max: p^T b$$

$$\text{st: } p^T A_j \leq 0, \quad j \in J$$

$$p \leq 1$$

p_i free

Note: c
is not present.

$$\text{If } RP > 0 \Rightarrow \text{DRP} > 0$$

Let $\bar{p}$ be dual sol'n.

Updating the Dual Solution

$\bar{p}$ is to DRP

$$p_{\text{new}} = p + \theta \bar{p}, \quad \theta \geq 0.$$

Choose θ so that: $(p + \theta \bar{p})^T A \leq c^T$

For $j \in J$, $\underline{p^T A_j = c_j}$, but: $\bar{p}^T A_j \leq 0$. / so all θ good.

For $j \notin J$: $\underline{p^T A_j < c_j}$. If $\underline{\bar{p}^T A_j > 0}$

then we have a bd on θ .

$$\theta: \min_{\substack{j \in J^c \\ \bar{p}^T A_j > 0}} \left[\frac{c_j - \bar{p}^T A_j}{\bar{p}^T A_j} \right]$$

This improves dual:

$$\begin{aligned} p_{\text{new}}^T b &= (p + \theta \bar{p})^T b \\ &= \underline{p^T b} + \theta \underline{\bar{p}^T b} \end{aligned}$$